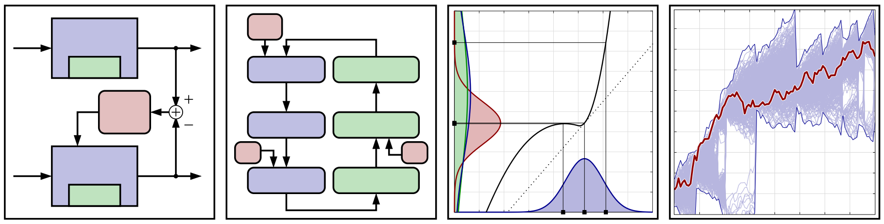
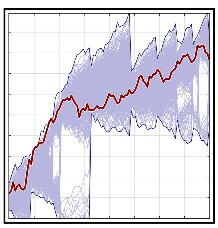
Welcome to the course!

- Welcome to ***Particle Filters!***
- This is the fourth course in a specialization that investigates the design and application of Kalman filters.
- This course focuses on generalizing the Kalman filter to problems requiring estimating the state of significantly nonlinear systems.



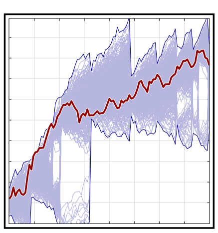
What topics will we study in this course?



In this course, you will learn:

- Background concepts relating to the goal of particle filters—to estimate the conditional pdf $f(x_k \mid \mathbb{Z}_k)$ accurately and efficiently.
- A brute-force method to estimate $f(x_k \mid \mathbb{Z}_k)$ that is computationally too complex for most applications.
- How to estimate integrals efficiently (via Monte–Carlo method) as a step toward simplifying the computation of $f(x_k \mid \mathbb{Z}_k)$.
- How to derive the particle filter using Monte–Carlo integration as its basis, and a significant problem with this approach.
- How to correct this problem to derive a robust implementation of particle filters.
- How to apply particle filters to assist indoor navigation.

What skills will you gain in this course?



After completing the course, you'll be able to:

- Execute provided Octave scripts that implement a direct (but inefficient) brute-force method for computing $f(x_k \mid \mathbb{Z}_k)$ and evaluate results.
- Execute provided Octave scripts that implement Monte–Carlo integration methods (with and without importance density) and evaluate results.
- Execute provided Octave scripts that implement the basic particle-filter algorithm and evaluate results.
- Execute provided Octave scripts that implement the robust particle-filter algorithm and evaluate results.
- Execute provided Octave scripts to investigate indoor navigation.

Prerequisites; for further study

Prerequisites:

- Course 1: ***Kalman Filter Boot Camp***
  - Basic background in KF concepts, random variables and random processes.
- Course 2: ***Linear Kalman Filter Deep Dive***
  - Derivation of the linear KF, and practice with math involving expectations.
- Course 3: ***Nonlinear Kalman Filters***
  - Notation describing nonlinear systems and derivation of nonlinear Kalman filters
- Lesson 4.1.2 briefly reviews necessary key prerequisite concepts and definitions.

Resource:

- I have learned a lot from Prof. James McNames of Portland State University.
- He has an excellent set of YouTube videos on “State Space Tracking” (including particle filters) at: `https://www.youtube.com/@JamesMcNames/videos`.
- For further study, you may wish to confer this optional resource.